

Richard
Stout

Thirteenth Annual In-House Convention 28.

29th April 1987

Room 4258/63

Department of Psychology

Dalhousie University

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Schedule

8:45 - 10:30

Session I Chair: V. LoLordo

1. J. Barressi
2. R. Marois, B. Chaisson, R. Croll
3. B. Chaisson, R.P. Croll
4. R. Croll
5. S. Bryson
6. A.R. Delamater, V. LoLordo

COFFEE BREAK

10:45 - 12:30

Session II Chair: J. Connolly

7. M. Harrington
8. V. Honig
9. W. Jia, I. Meinertzhagen
10. K. Briand
11. J. Enns
12. D. Phillips, S. Hall

1230 - 145

LUNCH IN LOUNGE

+

POSTER SESSION

13. J. McLaren, S. Bryson
14. R. Klein
15. J. Connolly
16. J. McLaren, R. Brown

1:45 - 3:30

Session III Chair: A. Droungas

17. P. McLeod, J. Mencher, J. Ryon, J.C. Fentress
18. T. Jukes
19. S. Marlin, M. Cynader
20. S. Hasan, S. Marlin, M. Cynader
21. P. McCormick
22. G. Prusky

COFFEE BREAK

3:45 - 5:30

Session IV Chair: R. Klein

- 23. R. Brown, S. Holman
24. C. Reed-Elder, V. LoLordo
poster- 25. R. O'Shea
26. M. Brennan
27. J. Fentress
28. D. Mitchell, M. Cynader

BEER + DINNER IN LOUNGE

ABSTRACTS

1. J. Barresi

Is Psychology an 'etic' or 'emic' science? And what of Biology?

An 'etic' science (e.g. 'phonetics') takes as its starting point a set of categories that are to apply to the phenomena of interest based on Cartesian principles of a hypothetico - deductive science. An 'emic' science (e.g. 'phonemics') applies categories that are grounded in the mental life of the subject of study, whether it be a man, a culture, or another organism. Its task is to rationally reconstruct that mental life in terms of rules that apply to mental activity in general. I will argue that the interests of these two types of science differ, and that psychology tries, unsuccessfully, to wed the two approaches.

2. R. Marois, B.J. Chiasson, R.P. Croll

Embryonic Development of Serotonin-Like Immunoreactivity in the Snail, Lymnaea

Modern immunohistochemical techniques allow for the identification of subpopulations of cells with distinct biochemical phenotypes within the central nervous system (CNS). The use of these techniques also allows for descriptions of the number, sizes and some aspects of the morphology of these cells. In this study we have exploited these advantages to identify the approximately 250 serotonergic cells found within the CNS of the pond snail, Lymnaea stagnalis at the time of hatching and then describe aspects of their embryonic development.

3. B.J. Chiasson, R.P. Croll

Post-hatching Development of Serotonin-Like Immunoreactivity in the Snail, Lymnaea

Post-hatching development of Lymnaea involves a 10-20 fold increase in the linear dimensions of ganglia comprising the central nervous system (CNS). As a first step toward an explanation of this growth we have examined the post-hatching development of a subpopulation of neurons which exhibit serotonin-like immunoreactivity. Growth of the CNS can be partially explained in terms of increases in both cell size and cell number within this subpopulation. Several large, identified neurons increase greatly in soma cross-sectional diameter. A large number of cells which are identifiable only as members of clusters tend to be smaller throughout development and tend to increase more slowly in cross-sectional diameter. In addition to these increases in cell size the population numbers of most cell clusters increase throughout development.

4. R.P. Croll

Distribution of Monoamines in the Hatchling Land Snail, Achatina Fulica

As was shown in the previous two presentations histochemical procedures can be used to describe single cells or clusters of cells within the nervous system of certain invertebrates. These techniques can also serve as a basis

for the tentative identification of cellular homologies between species. The glyoxylic acid histofluorescent method for the demonstration of monoamines and immunohistochemical procedures for the localization of serotonin were applied to the central ganglia of the land snail, Achatina fulica. Based on cell body size and location, biochemical content and neuritic projection patterns, several cells in Achatina appear to be homologous to cells previously identified in Lymnaea and several other gastropods.

5. S. Bryson

Preliminary Report on an Epidemiological Study of Autistic Syndromes

Autism is a stable disorder beginning in early childhood, and characterized by severe impairment in cognitive, linguistic and social development. The diagnosis of this syndrome has been the source of considerable controversy and confusion. The study to be described is an in-depth epidemiological investigation of the full spectrum of autistic conditions. In addition to providing the first North American prevalence data, we are examining the relationships among a number of psychological and physical variables. The aim is to provide data relevant to the aetiology of autism. This talk will focus on findings regarding genetic and traumatic contributory factors.

6. A.R. Delamater, V.M. LoLordo

Reinforcer location effects in simple conditioning: How does the rat know when and where to go to pick up food?

Two experiments inquire as to how the rat is sensitive to manipulations that vary the location of a reinforcer during simple conditioning procedures. In the first, we asked whether changing the location of a preferred edible after the rat had already learned that a stimulus signalled its availability in another place, would affect responses conditioned to that stimulus. Following the location-shift manipulation, conditioned responses continued to be elicited by the stimulus but they were directed to the shifted location. In the second experiment, we asked whether presenting the preferred edible in one location before the rat had learned that a stimulus signalled its availability in a different location, would affect the course of learning accruing to that stimulus. Preliminary data indicate that the rate of learning was enhanced in the location-shifted group relative to a non-shifted group. It is suggested that these data pose problems for most "reflex-oriented" conceptions of behavior.

7. M. Harrington

The mystery of the misplaced clock

Several laboratories have now reported that circadian rhythmicity, disrupted after ablation of the suprachiasmatic nucleus (SCN), can be restored by transplantation of fetal hypothalamic tissue containing SCN neurons. Transplant tissue has usually been placed into the part of the brain where the SCN is normally located - the base of the third ventricle. In collaboration with Dr. P. DeCoursey, I tested whether SCN tissue transplanted to a site distal to the hypothalamus would be capable of restoring circadian rhythmicity in hamsters made arrhythmic by SCN lesions.

8. W.K. Honig

The pigeon and the polka dot

Pigeons were trained to discriminate among matrices of 36 randomly arranged dots, some of them red, and the rest blue. If a matrix with red was positive, then the birds provided an orderly declining gradient of generalization as the proportion of red dots decreased. If blue was positive, the opposite gradient was obtained. During a transfer test with matrices of 64 dots the pigeons provided similar gradients according to the same ratios of red and blue as those that controlled the behavior with 36 dots.

9. W-G. Jia, I. Meinertzhagen

The Painless Tale of a Tunicate Headache
Identification of specific neurons in the larval tunicate

The larval ascidian (tunicate) has some vertebrate affinities and its nervous system contains relatively few neurons (about 50). This preparation provides a unique opportunity to study the development of nervous systems. In the present investigation, an acetylcholinesterase (AChE)-selective staining method and a methionine (MET)-enkephalin immunocytochemical technique was utilised to examine the nervous system in the larvae of Ciona intestinalis. Eight-10 primary sensory neurons and a pair of neurons in the vesical ganglion were identified by AChE staining and a single caudal neuron in the tail nerve cord showed MET-enkephalin immunoreactivity. This is the first demonstration of AChE-positive and MET-enkephalin reactive neurons in the larval tunicate, and the latter is also the first evidence for neuronal perikarya in the nerve cord.

10. K. Briand

The shape of things to come: Spotlight failure in exogenous covert orienting

A briefly presented visual stimulus shown in the periphery, is assumed to orient attention automatically. In fact, dissociations have been demonstrated between the effects of visual-spatial attention thus oriented, and attentional orienting which is more voluntary in nature. One rule of thumb which has been adopted is that a peripheral cue will enhance the processing of any possible item occurring at the cued location. The studies I report examine the alternative possibility that a peripheral cue, in fact, enhances processing of subsequent stimuli only to the extent that they are somehow physically similar to that cue. The rationale will be discussed, and preliminary data presented.

11. J. Enns

Does Life Imitate Art?

The development of perspective in children's drawings will be examined with an eye on the development of perspective in the history of Western art. Several experiments that examine one aspect of perspective in children's drawings -- how to draw occluded objects -- will be reported.

12. D.P. Phillips, S.E. Hall

Coding of Small Amplitude - Modulations by Cat Auditory Cortex Neurons

As part of an on-going research program on the neural coding of complex sounds, we have examined the responses of cat cortical cells to small (6-dB) linear modulations of the amplitude of continuous carrier tones. We found that these neurons were sensitive to the direction and rate of intensity change, and to the carrier level on which those modulations were imposed. Moreover, the fashion in which neural response rates varied with alterations in any of those stimulus parameters was of the kind predicted by our recent model of the afferent input to cortical cells. In conjunction with our previous evidence on the sensitivity of cortical neurons to other complex sounds, notably frequency modulations and tone-noise combinations, these data reinforce our view that the auditory cortex contains much or all of the neural machinery required for the analysis of sounds with complex spectral and temporal microstructure, including speech.

13. J. McLaren, S. Bryson

A Review of Recent Epidemiological Studies of Mental Retardation: Canadian and Non-Canadian Data on Prevalence, Associated Disorders and Aetiology

This poster will review recent epidemiological studies of mental retardation. Discussion will focus on Canadian versus non-Canadian data pertaining to 1) prevalence, 2) associated disorders, and 3) aetiology. The prevalence of mental retardation has been shown to vary as a function of degree of retardation, chronological age, and gender. Moreover, a large proportion of mentally retarded individuals present with additional disorders (e.g., language problems, behavioural difficulties, epilepsy, and cerebral palsy). The number of disorders varies with the degree of mental retardation. Investigations into the aetiology of mental retardation have focussed on the contributions of pre-, peri- and post-natal factors. Although it has traditionally been assumed that mild retardation (IQ=50-70) has a cultural familial basis, recent data have challenged this claim. The aetiologies of mild and severe (IQ<50) retardation will therefore be compared. The limitations of current epidemiological data and implications for future research will be discussed.

14. R. Klein

Does "inhibition of return" operate in visual search?

After a peripheral event the processing of nearby stimuli is facilitated for about 300 msec. From about 300 msec onwards there is a longer lasting inhibitory effect that has been called "inhibition of return". We conducted two experiments to determine if "inhibition of return" might operate to make visual search more efficient. Our main hypothesis was that when visual search appears to be serial and self-terminating (i.e. attention is inspecting each location), then inhibition would be left behind at each inspected location. We used a dual task in which the search response was followed immediately by a simple luminance detection probe that was presented "on" a potentially inhibited location or on some other location ("off"). Using Treisman's "popout" condition as a control for most visual

variables we found evidence consistent with a functional role for inhibition of return in visual search.

15. J.F. Connolly, D.W. Scott

Peripheral filtering of unattended input revisited: A brainstem auditory evoked response study.

Psychological theories of selective attention have been particularly concerned with the degree to which unattended inputs are processed. Proposals have ranged from peripheral filtering of unattended input to complete central processing of all input before the initiation of any selective processes. Attempts to determine which of these theories is correct and the nature of the physiological mechanisms involved have produced conflicting results. Experimentation in non human species has demonstrated attention driven efferent modulation of brainstem (olivocochlear bundle) responses to auditory input. Attempts to duplicate these findings in humans have resulted in a good deal of controversy with claims of both success and failure. In a recent study (Connolly, Aubry and McGillivray, 1987) the conditions associated with peripheral filtering in animals were adapted to human experimentation. No evidence was found in the brainstem auditory evoked response (BAER) for attention related response modulation. This failure to support the idea of peripheral filtering of unattended input in humans could not be considered definitive until greater task demands were introduced into the design. We report here BAER amplitude and latency data collected when subjects were required to attend and respond to: 1. the auditory stimuli evoking the response; 2. visual stimuli in conditions where the concurrent auditory stimuli assisted performance; 3. visual stimuli where the auditory stimuli hindered performance; and 4. visual stimuli where the auditory stimuli were of little consequence for performance. Results are discussed in terms of their implications for cognitive theories of selective attention and the putative physiological mechanisms mediating the process.

16. J. McLaren, R. Brown

Child Abuse: Behavioral and Developmental Signs

Children often present with problems such as somatic complaints or learning disorders which may mask abuse. Although maltreatment is frequently not identified as a causal or contributing factor, there may be signs in these children which indicate that abuse should be considered as a differential diagnosis. This poster reviews the behavioral and developmental signs which have been observed in physically abused, neglected and sexually abused children. The problems most frequently associated with physical abuse are aggressive behavior, social maladjustment, low self esteem and developmental delays. Neglected children are often significantly delayed in mental, motor, language and social development. Sexually abused children are most commonly reported to be depressed, to commit or attempt suicide and/or to behave promiscuously. Some methodological considerations concerning the studies reviewed and suggestions for future research are discussed.

17. P.J. McLeod, J. Mencher, J. Ryon, J.C. Fentress

Why do dogs roll in stuff that stinks?

Despite the widespread occurrence of scent-rubbing behaviour in mammals, neither the factors responsible for eliciting this behaviour nor the function(s) it serves are understood. The question we address here is which characteristics of an odour are important in eliciting rubbing. To determine this, 14 odors in 7 classes were presented to groups of wolves. Their responses to these stimuli were analyzed using multidimensional scaling (MDS) techniques to illustrate how the wolves themselves were categorizing the odors. Differences in rubbing behaviour among individuals were also analyzed using MDS to examine the possible role of social factors in scent-rubbing behaviour.

18. T. Jukes

A methodology for explanation in the social sciences

Explanation of action involves finding and modelling the generative mechanisms which produce manifest behaviours. The method employed in the natural sciences is appropriate for the social sciences: Positivism's attempt to limit knowledge only to accessible mechanisms has stunted explanation in the social sciences. Underlying intentional and meaningful action are intrinsic and extrinsic generative mechanisms, both of which are necessary for a complete explanation of action. To produce models of these forces, the developmental methodology, as employed by Piaget and Vygotsky, is valuable. In overcoming constraints, the organism is seen in change, thus showing the generative mechanisms in action, effecting behaviour.

19. S.G. Marlin, M.S. Cynader

Of Holes in Space and Shifts in Time

In the cosmic space-time continuum there are billions and billions of stars. There are more cells in the brain than there are stars in the sky. Each neuron in our visual cortex views one small place in visual space. Light from distant stars must travel through space until after some fixed amount of time, it reaches our eyes. But the long journey is not over for that signal must then travel through many synaptic connections until it is registered in our visual cortex. The time necessary for this synaptic trek appears to be a constant (40-60 ms). We have investigated the effect of repeated stimulation of small portions of the receptive fields of neurons in area 17 of the cat visual cortex. We examined the changes in the spatial properties of these receptive fields and the temporal characteristics of the cells. We have found that if a small part of the receptive field undergoes adaptation, the following four events may occur 1) no adaptation; 2) general adaptation of the entire receptive field; 3) a decrease in response to stimulation only at the point of adaptation; or 4) a decrease at the point of adaptation as well as increased responses at adjacent points in the receptive field. We also examined the cell's latency to respond to see if these changes in the spatial properties of the receptive fields could result from making the afferent neurons more sluggish.

20. S.J. Hasan, S.G. Marlin, M.S. Cynader

Keeping an Eye on Your Other

Studies of individual neurons in visual cortex have demonstrated selective adaptation of neuronal responses to such stimulus parameters as direction of motion, orientation and spatial frequency. We have examined interocular transfer of some of these adaptive effects in neurons displaying different strengths of excitatory input from the two eyes. An optimally oriented high contrast square wave grating, selected to evoke vigorous responses from a given unit, was swept across the receptive field. Unit responses for both preferred and null directions of grating movement in each eye were determined prior to and following one minute of unidirectional stimulation. Following adaptation, most units showed decreased responsivity to stimulation in both the dominant and non-dominant eyes. A clear minority of cells, however, showed adaptation effects specific to the eye which had been stimulated. The degree to which direction selective adaptation effects transferred interocularly was unrelated to the cell's ocular dominance categories as measured by Hubel and Wiesel's seven point scale. Both simple and complex cells exhibited comparable amounts of interocular transfer of adaptation. A large fraction of the neurons studied exhibited interocular transfer of direction selective adaptation. However, direction selective adaptation could occur without obvious interocular transfer. In addition, interocular transfer could occur even when prolonged stimulation in one direction resulted in response decrement in both directions of stimulation. Since the motion aftereffect displays marked interocular transfer in humans, it is possible that cells like those exhibiting interocular transfer of direction selective adaptation play a significant role in this illusion. The further characteristics of these cells are currently under investigation.

21. P. McCormick

Midlocation placement strategy in the orienting of visual attention: A spotlight on the hemifield activation hypothesis

Previous research has shown that we are able to orient visual attention to spatial locations without moving our eyes. How is this visual attention distributed? Posner, Snyder & Davidson (1978) have proposed a spotlight metaphor describing a spatially restrictive phenomenon. In contrast, Hughes & Zimba (1985) have presented evidence suggesting that attention is allocated to the entire left or right visual half-fields. However, the Hughes and Zimba data could have been generated by a midlocation placement strategy (MPS) of an attentional 'beam'. Evidence against the hemifield activation hypothesis and current research on MPS is discussed.

22. G. Prusky

Nick: a teen with terminal foresight

We have characterised and localised a population of nicotinic acetylcholine (ACh) receptors in the cat visual cortex, superior colliculus (SC) and laterageniculate nucleus (LGN). In visual cortex, the distribution of these receptors is associated with the pattern of LGN termination. The receptor distribution in the SC and LGN closely resembles the pattern of retinal

termination. Surgically undercutting the visual cortex eliminates receptors in this area, however they remain intact when cortical cells are selectively removed. Localised, unilateral lesions of the LGN abolishes nicotinic receptors in the areas of the visual cortex corresponding to the LGN lesion area. Bilateral eye removal eliminates nicotinic receptors in the SC of both tecta, and unilateral enucleation selectively destroys these receptors in the SC contralateral to the ablation. In the LGN, bilateral enucleations diminish nicotinic receptors in all retinal projection areas, and unilateral ablations reduce the number of receptors only in the areas receiving an afferent projection from the previously intact eye. Taken together, these data describe the affinity of nicotinic ACh receptors for primary sensory afferent terminals in the SC and LGN, and LGN terminals in the visual cortex. These receptors, via this strategic location, provide a mechanism by which ACh can modulate transmitter release in the visual system.

23. R. Brown, S. Holman

Love and hate among the gerbils

We examined sexual behaviour and odour preferences in mated pairs of gerbils. They were given sexual behaviour tests with their own mate and with a novel partner. We predicted that gerbils would show more sexual behaviour and less aggression to familiar partners. While many of the behaviours showed this trend, the differences were not significant. Steve Holman has been repeating this experiment and I hope he has significant results by now. In a second experiment we examined responses of gerbils housed in mixed- or same-sex pairs to the urine odours of their cage-mates and novel gerbils. Gerbils housed in mixed-sex pairs approached and investigated the odour of their cage-mate and produced more modulated ultrasounds in the presence of this odour than in the presence of the odour of a novel gerbil. Gerbils housed in same sex pairs showed no such differential responsiveness to the odour of their cage-mates. Odour may be used for individual recognition and function to reduce aggression in social interactions between mated pairs of gerbils.

24. C. Reed-Elder, V.M. LoLordo

Comparison of pavlovian conditioned inhibition procedures having different temporal arrangements of excitor and inhibitor on nonreinforced trials

Pigeons were trained with simultaneous (Sim) or serial (Ser) Pavlovian conditioned inhibition procedures. Transfer of inhibition was tested. In the first test, the inhibitor was presented in various temporal arrangements with the training excitor (transfer across procedures). In the second test, the inhibitor was presented in compound with a second excitor (transfer across excitors). Inhibitors conditioned with Ser procedures suppressed responding both Sim and Ser trials but did not suppress responding to another excitor. Inhibitors established with a Sim procedure transferred across excitors but showed poor transfer to Ser procedures.

25. R. O'Shea

Orientation tuning and binocularity of binocular-rivalry adaptation in human vision

Subjects viewed adaptation stimuli for 1 minute. Then two rival test gratings, differing in orientation from each other by 90°, were presented one to each eye, for 1 minute and subjects recorded binocular rivalry. When one eye was adapted, rivalry alternations were briefer for adaptation and test gratings of similar orientation, especially in the adapted eye. However, when both eyes were adapted by rival gratings, the duration of rivalry alternations increased in both eyes equally, to produce symmetrical U-shaped orientation functions. This implies that rivalry adaptation affects only binocular units in the human visual system.

26. M. Brennan

Individual Differences in Attention: Type A and Type B and Cognitive Failure

A significant negative correlation was found between two self-report measures: the JAS (Jenkins Activity Survey) and the CFQ (Cognitive Failure Questionnaire). The JAS is used to distinguish Type A from Type B individuals. The CFQ assesses a person's absent-mindedness and everyday errors. Thus, the correlation showed that Type A's reported low CFQ and Type B's reported high CFQ. Previous literature has shown that (1) Type A and Type B individuals differ in their attentional mechanisms, and (2) high CFQ and low CFQ differ in attention. My research will involve looking at whether or not these attentional differences really exist.

27. J. Fentress

From Moscow with Love

My plan in this talk is to discuss some recent data on contextual controls in integrated movement (with K. Berridge: rats) in the still broader context of the March, 1987, Moscow meeting on movement. The key problems from which details will be selected are: a) how do we deal with separable action properties that are also connected to one another; b) how do we deal with movement dynamics and also stabilities in action; c) how do Halifax rats relate to Russian polit bureaucrats? Scenic slides as well as data (I hope!).

28. D.E. Mitchell, M. Cynader

The effectiveness of part-time reverse occlusion as a means for promoting recovery from the physiological effects of monocular deprivation in the kitten visual cortex

Previously it has been shown that certain regimens of part-time reverse occlusion can promote recovery of normal visual function in the deprived eye of kittens that had been monocularly deprived to 6 or 8 weeks of age. We now report on the extent of the physiological recovery in the visual cortex of four such kittens that had recovered normal vision in their deprived eye.

Considerable shifts of ocular dominance towards the deprived eye were observed in one of the two conditions of occlusion. Moreover, a number of cells exhibited tuning for retinal disparity that was of sufficient magnitude to provide a possible substrate for stereopsis.